Voltage-Summation-Based Compute-in-Memory SRAM Macro With Input-Bit and Synaptic-Bit Slicing

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***Abstract*—Compute-in-memory (CIM) technology has emerged as a promising solution for energy-efficient matrix-vector multiplication (MVM), addressing the inherent limitations of von Neumann architecture. However, conventional Kirchhoff's current law based CIM technologies suffer from drawbacks such as *I-R* drop and energy inefficiency due to static current flow over a wide dynamic range. This paper introduces an energy-efficient and *I-R* drop-free CIM architecture based on capacitive coupling principle. Synaptic weights represented in 4-bit two's complement format are stored in SRAM-based 9T1C bit-cells. The architecture utilizes input-bit slicing method to perform MVM operations without the need for multi-bit digital-to-analog converters (DACs), significantly improving energy efficiency. A neuron circuit employs successive integration and rescaling (SIR) process to handle input-bit slicing encoded analog MVM results efficiently. The fabricated 16 kb, 28-nm prototype chip achieves an energy efficiency of 230 TOPS/W and a throughput of 32.19 GOPS for MVM operation with 4-bit resolution input and synaptic weights at a 100 MHz clock frequency. Even with ADC-induced analog variability, the proposed architecture achieves 90.64% and 97.99% inference accuracy for single-layer and multi-layer networks on the MNIST dataset, and 86.25% on CIFAR-10, validating its effectiveness and practical viability for low-power edge inference in complex workloads.**

***Index Terms*—Capacitive coupling, compute-in memory, hardware accelerator, matrix-vector multiplication, SRAM.**

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I. Introduction

As GENERATIVE artificial intelligence (AI) technology advances, demand for energy-efficient AI accelerators continues to grow. Multiply-and- accumulate (MAC) operation, a fundamental operation in deep neural networks (DNNs), is heavily memory-access-intensive [1]. However, von Neumann architecture, which requires continuous data transfer between arithmetic logic unit (ALU) and memory unit across a bandwidth-limited bus, poses significant constraints on energy efficiency and operation speed, commonly known as the von Neumann bottleneck [2].

Compute-in-memory (CIM) technology has emerged as a promising solution to the von Neumann bottleneck by enabling in-memory matrix-vector multiplication (MVM), thus avoiding external bus limitations [2]. CIM is primarily classified into current-summation-based (C-CIM) [3], [4], [5], [6], [7], [8], [9], [10], [11], [12], [13], [14] and voltage-summation-based (V-CIM) approaches [15], [16], [17], [18], [19], [20], [21], [22], [23], [24], [28]. C-CIM performs MVM by applying input voltages across word lines (WLs), converting them to currents through cell conductance, and summing these along bit lines (BLs) using Kirchhoff's current law (KCL). The entire MVM can be done in a single read operation, leveraging full parallelism. However, as array size grows, the increased summed current imposes wider dynamic range requirements on neuron circuits, and accuracy suffers due to I–R drops across columns [25], [26], [27].

In contrast, V-CIM uses capacitance to encode weights and voltage differences as inputs. It performs MVM via voltage summation at a floating node based on capacitive coupling (CCP), avoiding static current and mitigating I–R drop-induced errors. Moreover, it operates within a fixed voltage range (e.g., V_{SS} to V_{DD}), easing neuron design scalability. Nonetheless, SRAM-based V-CIM implementations face challenges: DACs and ADCs introduce notable energy and area overhead. MACC-SRAM [24] mitigates ADC cost through an adder-first architecture using a switched-capacitor adder and reduces DAC energy via a stepwise-charging/discharging input driver (SCD-IDR). Still, DAC-related overhead remains.

In this study, we propose a SRAM-based V-CIM macro that eliminates the need for multi-bit DACs while enabling multi-bit input activations through input-bit slicing technique

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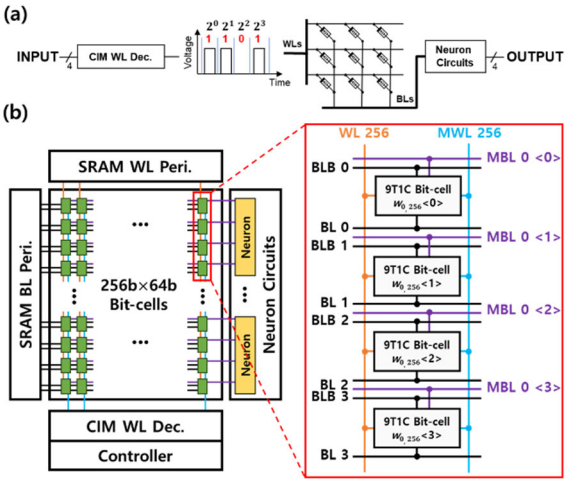


Fig. 1. (a) Data flow during CIM operation with input-bit slicing encoding method, and (b) architecture of the proposed V-CIM macro.

[29], [30], [31]. The fabricated V-CIM macro simultaneously supports input-bit slicing and synaptic-bit slicing, utilizing a neuron circuit based on a switched-capacitor adder with successive integration and rescaling (SIR). The V-CIM prototype chip, fabricated using TSMC 28 nm logic CMOS technology, achieves an energy efficiency of 230 tera-operations per second per watt (TOPS/W) under 0.9 V supply for 4b/4b weight/activation operations. It also attains a weight density of 485 kb/mm^2, representing one of the most advanced SRAM-CIM macro demonstrations in recent years.

II. Experiment

A. CIM System Using Capacitive Coupling Principle

To achieve high precision MAC operation, multi-bit representation of input, weight, and output is necessary. DACs are used for multi-bit input representation, and ADCs are used for digitizing output with multi-bit resolution. However, previous CIM approaches [20], [21], [22] report high power consumption and large area overhead due to ADCs and DACs.

The input-bit slicing method enables the representation of multi-bit resolution input using single-bit DACs. This novel method mitigates the need for multi-bit DACs, significantly enhancing energy efficiency while also reducing area overhead and system complexity. The method encodes input values as binary signals, where each time step corresponds to a specific input bit position, as depicted in Fig. 1(a). Following the input-bit slicing approach, the MVM operation is performed on a bit-wise basis, and the resulting bit-wise MVM outputs are accumulated within the neuron circuit. Upon completion of the final MVM operation, the bit-wise accumulated results are converted into a 4-bit digital output by the neuron circuits.

Fig. 1(b) presents the architecture of the proposed V-CIM macro. The V-CIM macro consists of a 256×64 bit-cell array, SRAM peripherals for read/write operations, CIM word-line (WL) decoder, neuron circuits, controllers, and bus interface. To realize multi-bit representation of synaptic weight, we adopted the synaptic-bit slicing method. A 4-bit two's complement synaptic weight is mapped across four distinct bit-cells sharing identical WL. A single neuron circuit processes MVM results in parallel across four rows of bit-cells, handling MAC operation result from 4-bit inputs and weights. This enables the V-CIM macro to achieve fully parallel analog MVM operations with 256 4-bit inputs and 256×16.4-bit two's complement weights.

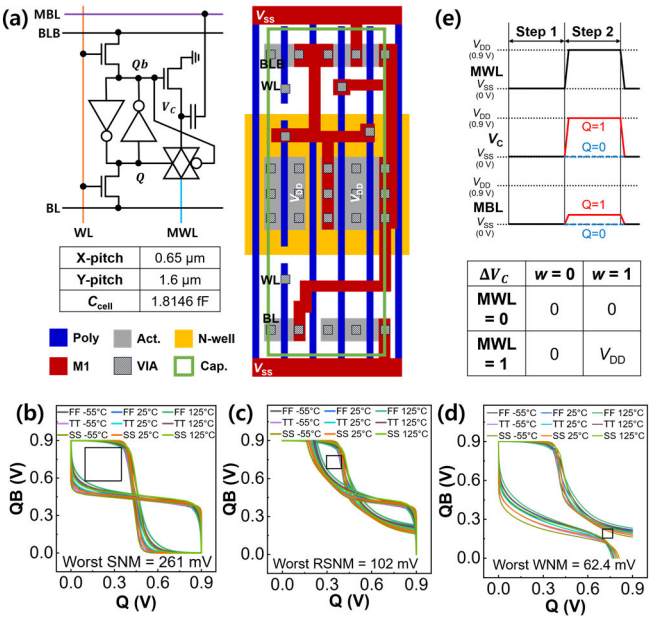


Fig. 2. (a) Schematic and layout of the proposed 9T1C bit-cell. Voltage transfer characteristic analysis of the 9T1C bit-cell: (b) static noise margin (SNM) butterfly curves, (c) read static noise margin (RSNM) butterfly curves, and (d) write noise margin (WNM) butterfly curves, simulated across TT, FF, and SS corners at −55 °C, 25 °C, and 125 °C. The worst-case SNM and RSNM occur at the FF corner under 125 °C, while the worst-case WNM occurs at the SS corner under 125 °C. (e) Operation method of 9T1C bit-cell.

B. Structure and Operation Method of Bit-Cell

Fig. 2(a) shows the schematic and layout of the 9T1C bit-cell [33] used in the proposed V-CIM macro. Compared to a standard 6T SRAM, this cell adds three transistors and a metal-oxide-metal (MOM) capacitor. It introduces dedicated ports—MAC WL (MWL) and MAC BL (MBL)—for CCP-based MAC operations. Each 4-bit two's complement weight is implemented using four identical 9T1C cells sharing the same WL. The MOM capacitor is placed above the 9T cell region, keeping the bit-cell area unchanged.

The MOM capacitor offers high PVT robustness. Global process shifts affect all cells similarly, preserving relative matching and minimizing errors during capacitive coupling. Additionally, its capacitance is stable over voltage and temperature, eliminating the need for external calibration.

The capacitor is sized at 1.8146 fF, maximizing signal strength within the layout constraints and mitigating parasitic effects on MBL. Bit-cell stability is evaluated through SNM, RSNM, and WNM, extracted via butterfly curves [34] based on voltage transfer characteristics (VTC). As shown in Fig. 2(b)–(d), the minimum values across PVT corners are 261 mV (SNM), 102 mV (RSNM), and 62.4 mV (WNM).

The minimum supply voltage ($V_{DD,min}$) is determined via 5,000-point Monte Carlo simulations under varying V_{DD}. Based on the criteria in [35], [36], and [37], $V_{DD,min}$ is defined where SNM and RSNM exceed thermal voltage ($\approx$ 26 mV at 300 K), and WNM > 0 mV. The resulting $V_{DD,min}$ is 684 mV, with RSNM being the limiting factor.

Once the MAC operation begins, the MBLs are first discharged to V_{SS} by the reset footers in the neuron circuit, as shown in Fig. 3(a). After the discharge phase, the reset

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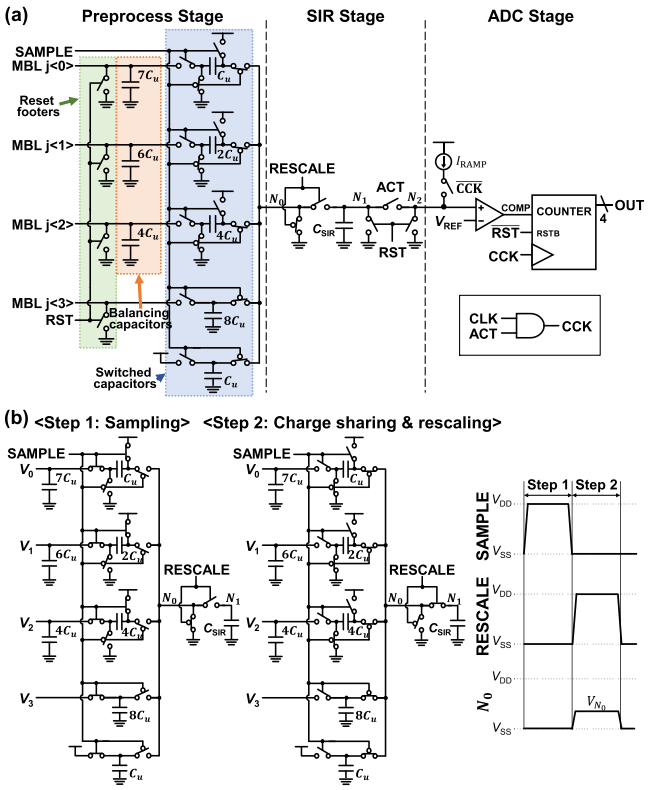


Fig. 3. (a) Schematic design of the proposed neuron circuit. CCK denotes the gated clock signal used for ramp ADC. (b) Step-by-step illustration of the preprocessing stage with the corresponding equivalent circuit. The preprocessing operation is realized through a sequence of switched-capacitor reconfigurations, as depicted in the timing diagram, which determines the initial voltage condition for subsequent neuron evaluation.

footers are turned off, leaving the MBLs in a floating state, with the reset footers being the only driving path during the initialization phase. During the subsequent MAC operation, inputs are applied to MWLs, and the computation results are accumulated on floating MBLs. Since analog MAC operation relies on the CCP, voltage at top plate of the capacitor (V_C) varies while the MBLs remain floating. In the V-CIM macro, the MBLs is discharged to V_{SS} and then left floating before beginning the MAC operation. Fig. 2(e) outlines the operation.

In Step 1, MWL is pulled to V_{SS}, setting the capacitor top plate voltage (V_C) to V_{SS}, independent of stored data. Inputs are represented as unsigned 4-bit integer (UINT4). This ensures that every input bit applied through the MWL always transitions from V_{SS} to V_{DD}.

In Step 2, input "1" causes MWL to swing to V_{DD}; "0" keeps it at V_{SS}. If the weight bit is 1, the MWL-side pass transistor is on, and V_C rises to V_{DD}. If the weight is 0, V_C stays unchanged due to the enabled V_{SS}-side pass transistor.

During the MAC operation, the BL and BLB voltages do not participate in the analog computation. The MAC result is determined solely by the stored bit-cell states (*Q*/*Qb*) and the applied MWL voltages through capacitive coupling, while BL/BLB remain statically biased and electrically decoupled from the accumulation path. The floating MBL voltage increases with the number of active bit-cells. For N contributing cells, the MBL voltage is given by:

$$V_{MBL} = \frac{\sum_{i=1}^{N} C_{cell} \times \Delta V_{C,i}}{\sum_{i=1}^{N} C_{cell} + C_p} \quad (1)$$

where $\Delta V_{C,i}$ s the capacitor top-plate swing of the i-th bit-cell during MAC operation, and C_p is the MBL's parasitic capacitance.

C. Structure and Operation Method of Neuron Circuit

The proposed V-CIM macro adopts both input-bit and synaptic-bit slicing, enabling analog MVM to be computed per input-bit position. The resulting four partial sums are sequentially conveyed to the neuron circuit through four MBLs at different time slots, each corresponding to a different input bit.

As shown in Fig. 3(a), the neuron consists of three stages: preprocess, SIR, and ADC. The preprocess stage interfaces with the four MBLs aligned to a synaptic row and combines their voltages into a single analog output. It includes switched capacitors, balancing capacitors, and reset footers. To ensure proper bit-wise weighting, the switched capacitors are configured as $8C_u$, $4C_u$, $2C_u$, and C_u from *MSB* to *LSB* (C_u: unit capacitance). To balance the varying load on each MBL, compensating capacitors (e.g., $7C_u$ for *LSB*, $4C_u$ for *MSB*z1) are added, maintaining uniform total capacitance across inputs. A footer discharges all MBLs to V_{SS} before each operation cycle.

Fig. 3(b) shows the operation of the preprocess stage. In Step 1, MBL voltages (V_0 to V_3) are sampled during the SAMPLE signal. In Step 2, charge sharing occurs, producing a weighted sum voltage at the node N_0:

$$V_{N_0} = \frac{V_{DD}}{2} + \frac{8V_3 - (1V_0 + 2V_1 + 4V_2)}{16} \quad (2)$$

Once the voltage at node $N_0$is established, the RESCALE signal initiates the SIR stage. The SIR capacitor (C_{SIR}) is pre-discharged to V_{SS} during initialization and then shares charge with node N_0. The C_{SIR} is carefully matched to the total capacitance at N_0, enabling the accumulated voltage to be halved with each new MVM result, which reflects the *LSB*-to-*MSB* bit significance.

The preprocessed partial sum from previous stage is accumulated in the SIR stage via input pulses. When the SAMPLE signal deasserted and the RESCALE signal is asserted, charge sharing occurs between nodes N_0 and N_1. This recursive integration process yields:

$$V_{C_{SIR}}(k) = \frac{V_{N_0}(k) + V_{C_{SIR}}(k-1)}{2} \quad (3)$$

Here, $V_{C_{SIR}}(k)$ represents the accumulated partial MVM result up to the k-th input-bit cycle, with each cycle incorporating the appropriate binary weight through recursive halving. As the number of accumulation steps increases, the contribution of the previously accumulated $V_{C_{SIR}}$decreases. Therefore, the accumulation starts from the *LSB* cycle ($k = 0$) and proceeds toward the *MSB*.

This structure accumulates successive bit-sliced results with diminishing weight, emulating a 4-bit input resolution. After processing the *MSB*, the SIR capacitor voltage reflects the full MVM result.

The analog MVM result is then digitized by the ADC stage. While SAR or Flash ADCs could be used, they require amplifiers and buffers, leading to significant power and area overhead. To avoid this, a ramp ADC is employed for its simplicity and efficiency.

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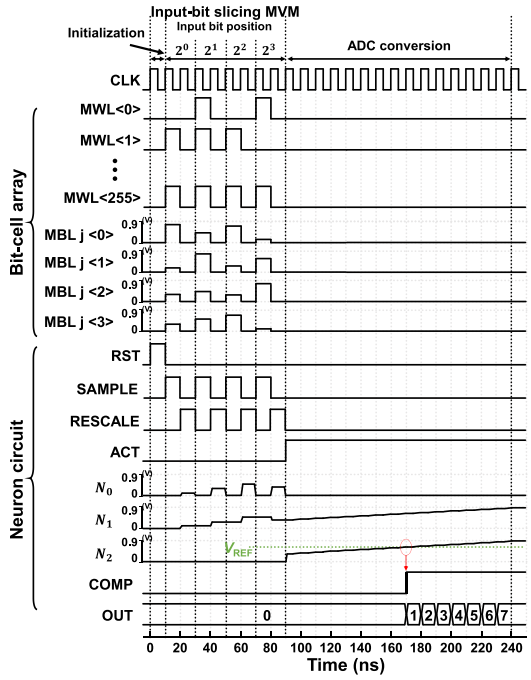


Fig. 4. Timing diagram of the V-CIM macro during CIM operation.

During ADC, the ACT signal is asserted and RESCALE is deactivated. A gated clock (CCK) drives the ramp, issuing 15 steps for 4-bit resolution. When the voltage at node N_2 exceeds the comparator's reference voltage (V_{REF}), a counter is triggered. By adjusting V_{REF}, the firing timing of the COMP signal can be controlled. This enables compensation for cases where the analog MAC result exhibits a variation-induced offset. After all CCK pulses, the final counter value corresponds to the digitized MVM result.

Fig. 4 presents the timing sequence. At the start of CIM, the input *LSB* is applied via MWL, synchronized with the SAMPLE pulse. With each subsequent SAMPLE falling edge, higher input bits are sequentially loaded into the bit-cell array. This continues until the *MSB* is processed. Based on this timing, a complete CIM operation, including MVM, RESCALE, and ADC, requires 24 clock cycles.

III. Result and Discussions

The V-CIM macro features a 256×64 array of bit-cells, corresponding to 256 MWLs and 16 groups of MBLs. It is capable of processing MVM operations with a 256-element input vector, where each value ranges from 0 to 15, and a 256×16 weight matrix, where each value ranges from -8 to 7. This configuration allows the macro to handle MAC results ranging from $-256 \times 15 \times 8$ to $256 \times 15 \times 7$ which is equivalent to a range of $-30{,}720$ to 26,880.

A. Post-Layout Simulation Analysis

Fig. 5(a) presents simulated waveforms corresponding to the timing diagram shown in Fig. 4. The simulation captures internal analog signal transitions during a single MAC operation, including MWL activation, voltage responses at internal neuron nodes (N_1, and N_2), and the generation of

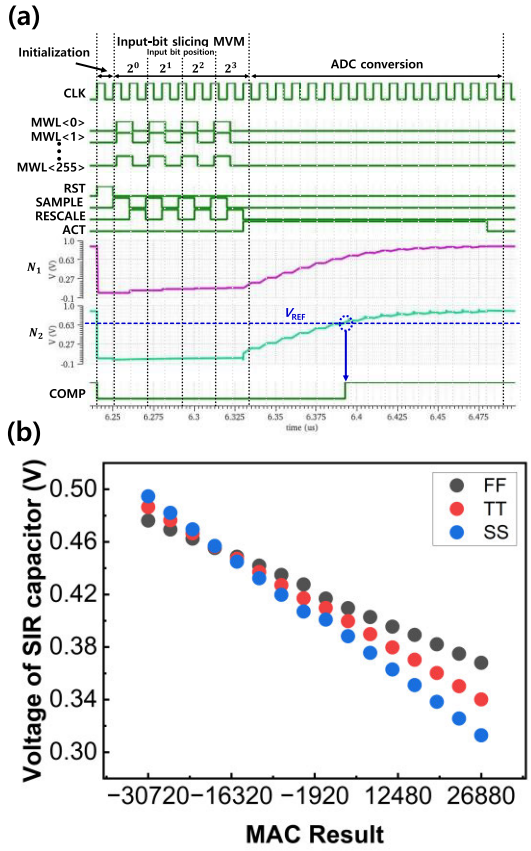


Fig. 5. (a) Simulated waveforms of the proposed V-CIM macro during a single MAC operation. (b) Voltage of the SIR capacitor of the neuron circuit in process corners: FF, TT and SS.

the comparator output. These results validate the functional correctness of the proposed timing scheme and demonstrate the feasibility of analog MVM operation using the V-CIM macro.

Fig. 5(b) illustrates the CIM operation results across three major process corners at 25°C: typical-typical (TT), fast-fast (FF), and slow-slow (SS). It depicts the analog voltage level at top plate node of the SIR capacitor (N_1) in the neuron circuit prior to ADC conversion. The results demonstrate a linear relationship between the MAC result and voltage of the SIR capacitor. Although process variations affect the slope of the SIR capacitor voltage, a linear ADC output can be achieved by adjusting the reference voltage (V_{REF}) and modifying the charging current of the ramp ADC.

B. Hardware Measurement

Fig. 6(a) shows the optical micrograph of the fabricated V-CIM prototype chip, implemented in TSMC 28 nm CMOS logic process. The chip integrates a 16 kb CIM array within a compact area of 0.033 mm^2. Control logic, including the CIM WL decoder, CIM and SRAM controllers, and bus interface, is grouped within the "CIM WL Dec. & Controllers" block.

Fig. 6(b) presents the area distribution. The CIM array accounts for 54.2% of the total area, while the decoder and control logic occupy 24.1%. The neuron circuits, despite integrating capacitors and ADCs per neuron, use only 7.3% of the area. This compact implementation is made possible by

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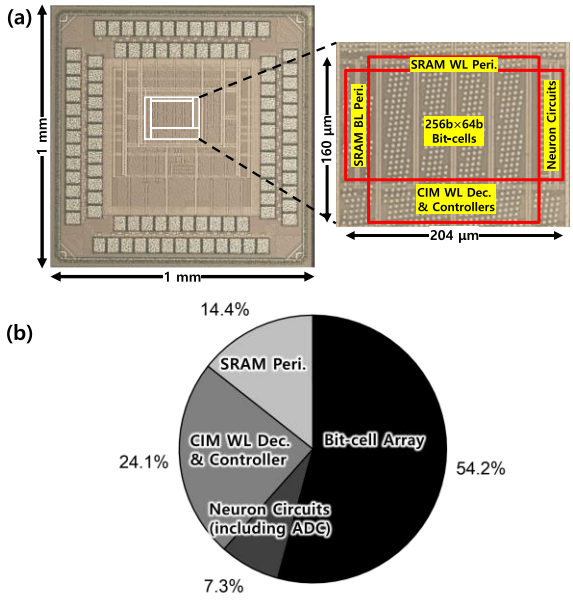


Fig. 6. (a) Optical micrograph and (b) area breakdown of the prototype V-CIM chip.

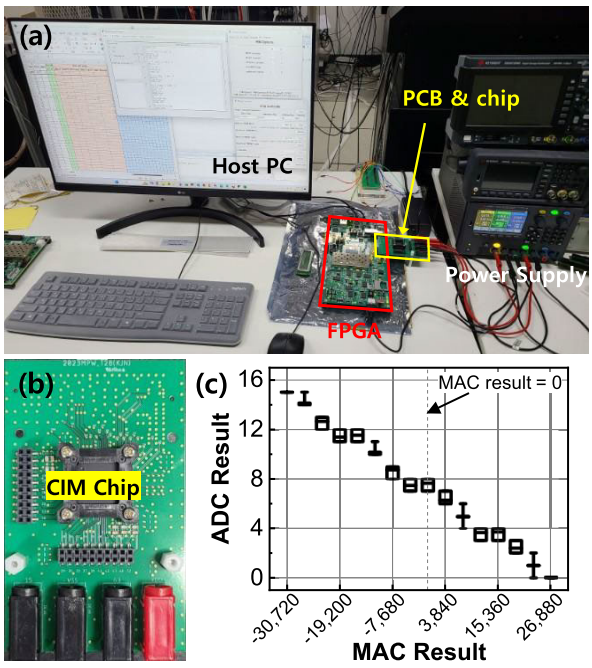


Fig. 7. (a) Hardware measurement setup and (b) PCB integrated prototype V-CIM chip. (c) Measured ADC result within various MAC result range.

using MOM capacitors, enabling high-density integration with minimal area overhead.

The digital test signals are generated by an FPGA board. Power and bias voltages for the prototype chip are supplied via external DC sources. As shown in Fig. 7(b), the chip is packaged and mounted on a PCB through a socket for evaluation. The PCB interfaces directly with the FPGA via an FMC connector, reducing parasitic inductance and ensuring efficient control and data transfer.

Fig. 7(c) shows measured hardware results across different MAC output levels. Each case involved 1,000 MVM operations using a single chip. The resulting box plots demonstrate

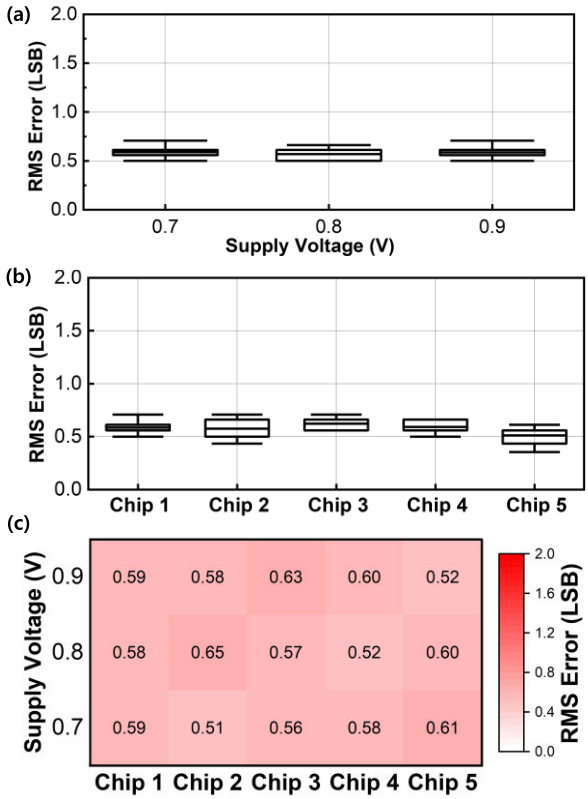


Fig. 8. Measured RMS error: (a) supply voltage dependence, (b) chip-to-chip variation, and (c) average over all conditions.

linear and consistent ADC behavior, confirming the higher MAC values result in lower ADC outputs as designed. The standard deviation of ADC outputs per MAC level was also extracted, with a maximum of 0.*61 LSB* and an average of 0.*52 LSB*. Despite minor localized non-linearities, the sub-*LSB* variations demonstrate repeatability and robustness—sufficient for approximate computing applications like neural network inference.

To assess the system-level impact, the measured variability was incorporated into neural network inference simulations. As discussed in Section III-C, the resulting accuracy remained largely unaffected, validating the system's tolerance to analog noise.

To assess system robustness under chip-to-chip and supply voltage variations, we measured the root-mean-square (RMS) error across different operating conditions. Fig. 8(a) and 8(b) show the RMS error trends with varying supply voltages and across multiple fabricated chips, respectively. Compensation was applied through tunable parameters I_{RAMP} and V_{REF} (see Fig. 3). As summarized in Fig. 8(c), the RMS error consistently remained below *1 LSB* in all cases, ranging from 0.51 to 0.*65 LSB*, demonstrating reliable and stable system behavior.

Fig. 9 shows the energy breakdown of the V-CIM macro during a single CIM operation at 100 MHz. Each operation computes 256×16 multiplies and 255×16 additions, totaling 8,176 operations. The total energy per operation is 35.61 pJ, distributed as follows: 22.9% for the CIM WL decoder, 32.3% for the bit-cell array, and 27.1% for the neuron circuits, resulting in 4.355 fJ per arithmetic operation.

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TABLE I
COMPARISON TO PRIOR CIM ARCHITECTURES

	This Work	JSSC'20 [18]	JSSC'23 [21]	TCAS I'21 [22]	ISSCC'20 [23]	JSSC'24 [24]	TCAS I'24 [39]	VLSI' 21 [40]
Computation Type	CCP	CCP	CCP	CCP	KCL	CCP	CCP	CCP
Technology (nm)	28	65	22 FFL	28 FDSOI	7	28	28	28
Macro size (kb)	16	16	128	16	4	16	64	288
Macro area (mm^2)	0.033	0.081	0.25	0.051	0.0032(4)	0.031	0.197	0.51
Supply voltage (V)	0.7/0.9	0.6 ~ 1	0.7/1.1	1	0.65 ~ 1	0.9	0.8	0.9/0.6
Input resolution (b)	4	1	8	5	4	4	8	5
Weight resolution (b)	4	1	8	2/4/8	4	4	8	1
ADC precision (b)	4	5	8	2 ~ 5	4	6	8	8
Clock frequency (MHz)	100	50	145/240	214	182	566	250	-
Macro Throughput (GOPS)	32.19	1,638	600/1,000	876.54(3)	372.4	211	256	6,144
Power consumption (μW)	66.4/140.2	2,439.3	18,634 /64,516	7,342	1,061	1,295	3,597	5,301
Energy efficiency, EE (TOPS/W)	485/230	671.5	32.2/15.5	119.38(3)	351(5)	163	71.17	1,159
Normalized EE(1) (TOPS/W)	7,760 /3,680	1,270	1,171/826	1,194	1,966	2,608	4,555	5,795
Area efficiency, AE ($TOPS/mm^2$)	0.975	20.22	2.4/4	17.19	116.4	6.81	1.299	2.4
Normalized AE(2) ($TOPS/mm^2$)	15.61	84.36	102.0/170. 0	171.9	177.4	108.9	83.14	12

(1) Normalized to 1-bit input and weight, Scaled to 28 nm, assuming 30% energy efficiency improvement per technology node. Normalized energy efficiency = (input bit width) × (weight bit width) × $(1+30\%)^{\log_2((\text{technology}^2/28^2))}$ × (TOPS/W) [21]
(2) Normalized to 1-bit input and weight, Scaled to 28 nm, assuming 80% area efficiency improvement per technology node. Normalized area efficiency = (input bit width) × (weight bit width) × $(1+80\%)^{\log_2((\text{technology}^2/28^2))}$ × ($TOPS/mm^2$) [21]
(3) The conditions are as follows: input resolution:5-bits, weight resolution: 2-bits, ADC precision 2-bits
(4) Including testing & reconfigurable blocks
(5) The average TOPS/W is applied

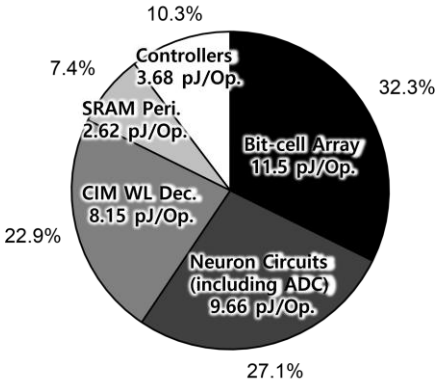


Fig. 9. Energy breakdown during a single CIM operation.

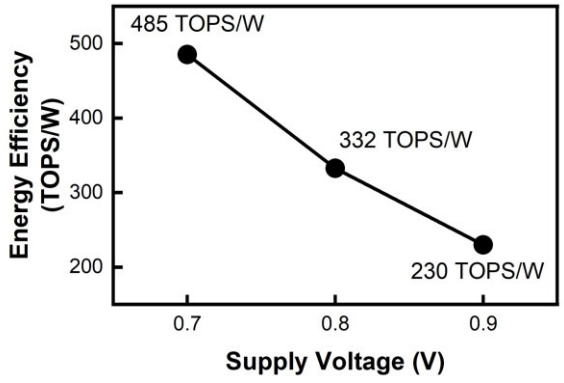


Fig. 10. Energy efficiency of the V-CIM macro at various supply voltages.

Prior works [20], [21], [22] report that DAC and ADC circuits account for a major portion of CIM energy—56% in [20], 85.5% in [21], and up to 70% in [22]—highlighting them as energy bottlenecks in analog CIMs. In contrast, the V-CIM macro reduces this burden by adopting CCP and input-bit slicing. Its WL decoder (acting as a DAC) and neuron circuits (ADC) jointly consume about 50% of the total energy, without requiring conventional multi-bit DACs or complex ADCs.

The macro performs parallel MACs between a 256-length input vector and a 256 × 16 weight matrix, completing one operation in 24 clock cycles. At 100 MHz, this translates to 32.19 GOPS and 230 TOPS/W energy efficiency.

To analyze voltage scaling, energy efficiency was measured at 0.7 V, 0.8 V, and 0.9 V. As shown in Fig. 10, lowering V_{DD} improves energy efficiency due to the quadratic nature of dynamic power consumption.

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In general, reducing V_{DD} degrades access speed due to reduced drive strength, which would require a longer clock period to maintain correct operation, leading to a trade-off between throughput and energy efficiency. However, in this work, the clock is externally supplied and fixed, and the evaluated voltage range is restricted to values above the minimum operating voltage ($V_{DD,min}$) required for reliable SRAM operation. Therefore, the observed improvement in TOPS/W primarily reflects the reduction in dynamic power without frequency scaling.Table I compares the proposed macro with existing designs, using normalized metrics based on [21]. At 0.9 V and 100 MHz, the normalized energy and area efficiencies are 3,680 TOPS/W and 15.61 TOPS/mm^2, respectively. At 0.7 V, energy efficiency improves to 7,760 TOPS/W.

While competitive in energy efficiency, the macro shows lower area efficiency due to the synaptic bit-slicing scheme. Each 4-bit weight uses four separate 9T1C cells, increasing area and lowering effective density. This reflects a trade-off between weight resolution and area efficiency in the proposed architecture.

When compared with prior CIM macros employing DAC-based input interfaces under the same 4-bit input resolution, the proposed design demonstrates a clear advantage in energy efficiency. Compared to the SRAM-based CIM macro reported in JSSC'23 [40], which supports 4-bit inputs and 8-bit weights using charge-sharing–based accumulation, the proposed macro achieves more than a 10× higher normalized energy efficiency (7,760 TOPS/W at 0.7 V versus 707 TOPS/W). Although the JSSC'23 design attains higher raw throughput, its energy efficiency is significantly influenced by the use of multi-bit analog input generation and associated peripheral overheads.

Similarly, compared to the SRAM-based CIM macro reported in SSCL'22 [41], which supports 4-bit inputs and 4-bit weights using CCP-based computation, the proposed macro exhibits approximately 9× higher normalized energy efficiency under the same input and weight precision. The SSCL'22 design relies on DAC-assisted multi-bit input representation, whereas the proposed macro directly processes digital input bits using an input-bit–sliced voltage-summation scheme.

These comparisons indicate that, under equivalent input resolution, eliminating the DAC at the input interface can significantly improve energy efficiency. The proposed architecture therefore provides a more energy-efficient and system-friendly solution for low-power on-device inference, where input precision is constrained and peripheral overhead dominates overall power consumption.

C. System-Level Evaluation With Hardware Measurements

To evaluate the applicability of the V-CIM architecture to neural network tasks, we performed system-level simulations based on hardware-measured ADC behavior. The MNIST dataset was used as a benchmark, with 28 × 28 grayscale images downscaled to 4-bit resolution to match the V-CIM format.

A fully connected (FC) neural network was constructed for classification. FP32-trained weights were quantized to INT4 via uniform quantization and mapped directly to the bit-cell array. The MAC outputs generated by the neuron circuit were interpreted through an activation function block. Since the neuron circuit outputs unsigned INT4 values (0–15), we applied a format transformation:

$$I(x) = 15 - (x - 8) \tag{4}$$

For ReLU activation, the following was used:

$$\mathrm{ReLU}(x) = \max(I(x), 0) \tag{5}$$

Using quantization-aware training (QAT), Model 1 (single-layer FC) achieved 91.33% accuracy, comparable to the baseline (91.45%). Model 2, a deeper FC network, achieved 98.04%, also close to its baseline (98.14%), as shown in Table II.

To evaluate the applicability of the proposed V-CIM macro to more complex workloads, we evaluated a CNN (Model 3) using the CIFAR-10 dataset, characterized by higher complexity due to its multi-channel color input. The CNN was structured following [18], with convolution and FC layers executed on the V-CIM macro. Supporting operations (e.g., pooling, reshaping) were assumed to run digitally. Weight-stationary mapping was applied: $3 \times 3 \times N$ convolution kernels were mapped as FC(9, N) operations.

Model 3 achieved 88.89% inference accuracy—close to the 90.77% software baseline—again trained using QAT under

TABLE II
ACCURACY COMPARISON

Model	Layers	Activation Functions	Baseline Accuracy	Chip Accuracy
1	*FC(784,10)	Identity	91.45%	91.33%*** (90.64%)****
2	FC(784,256)	ReLU	98.14%	98.04%*** (97.99%)****
	FC(256,64)	ReLU		
	FC(64,10)	Identity		
3	**Conv(3,128)	ReLU	90.77%	88.89%*** (86.25%)****
	Conv(128,128)	ReLU & Max pool		
	Conv(128,256)	ReLU		
	Conv(256,256)	ReLU & Max pool		
	Conv(256,512)	ReLU		
	Conv(512,512)	ReLU & Max pool		
	FC(8192,1024)	ReLU		
	FC(1024,1024)	ReLU		
	FC(1024,10)	ReLU		

* FC(m,n): m×n fully-connected layer
** Conv(m, n): 3×3 convolution layer with m input and n output channels
*** Accuracy evaluated using quantized input, weight, and activation values.
**** Accuracy evaluated with ADC conversion noise included.

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4-bit constraints, demonstrating the macro's effectiveness for higher-complexity workloads.

To evaluate analog robustness, measured ADC variations (mean 0.*52 LSB*, max 0.*61 LSB* from Section III-B) were modeled as Gaussian noise and injected before quantization during simulations. Accuracy degradations were minimal: Model 1 dropped by 0.69%, Model 2 by 0.05%, and Model 3 by 2.64% (from 88.89% to 86.25%), confirming stable performance under realistic analog variability.

IV. Conclusion

CIM technology offers a promising pathway for energy-efficient on-device AI acceleration, overcoming limitations of the traditional von Neumann architecture. Although C-CIM approaches mitigate data movement bottlenecks, they still encounter challenges such as I–R drop, high power consumption, and wide dynamic range requirements for output currents. In contrast, the proposed V-CIM architecture performs static-current-free MVM operations at floating nodes, addressing these issues effectively. By employing an input-bit slicing method, the design eliminates the need for multi-bit DACs during CIM operations, significantly improving energy efficiency. The neuron circuit processes the bit-sliced analog MVM results via capacitor-based charge sharing. This co-optimized input encoding and computation scheme forms the foundation of a highly efficient architecture. Fabricated in a 16 kb prototype using 28 nm CMOS, the proposed V-CIM macro achieves 230 TOPS/W energy efficiency and 32.19 GOPS throughput with 4-bit inputs and weights at 100 MHz.

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